



Department of Computer Science and Engineering

**COMP 6771 - Image Processing
Fall 2024**

Project Report - Seam Carving for Content-Aware Image Resizing

Submitted by:

**Anurag Teckchandani - 40263724
Jaiwant Singh Mahun - 40270569**

Name	Contributions
Anurag Teckchandani	• Compared with a recent research paper and analyzed differences in methods and results. • Tested algorithm with varying parameters and recorded results. • Compared results with baseline methods and prepared visualizations for report. • Finalized report, ensured compliance with format guidelines, and prepared submission files.
Jaiwant Singh Mahun	• Developed core algorithm logic and implemented core functions using OpenCV • Analyzed challenges and pros/cons of the implemented algorithm. • Validated the implementation using paper-provided and custom test images. • Drafted sections on paper review and methodology, including figures and flowcharts.

Part 1

Paper 1: Seam Carving for Content-Aware Image Resizing [1] (Candidate Paper)

Motivation and Contributions - The paper addresses the shortcomings of traditional resizing methods, like scaling and cropping, which distort or remove key content. The authors introduce seam carving, a content-aware technique that adapts resizing to an image's visual structure.

Key contributions include:

- Defining seam carving to remove or insert low-energy pixel paths, preserving critical features.
- Applications for resizing, retargeting, and object removal.
- Proposal of multi-size images for real-time, adaptive resizing.

Approach -

- **Energy Map Calculation:** Compute pixel importance using gradient magnitudes.
- **Seam Identification:** Use dynamic programming to find optimal seams of low-energy pixels.
- **Seam Manipulation:** Remove seams to shrink or duplicate them to enlarge the image.
- **Applications:** Enable content-aware resizing, object removal, and aspect ratio adjustments.

Critique -

Strengths: Innovative and effective for preserving important content during resizing. Also, Flexible applications, including object removal and retargeting.

Weaknesses: Struggles with complex images where multiple critical areas exist and lacks quantitative evaluation; primarily relies on visual examples.

Paper 2: CarvingNet: Content-Guided Seam Carving Using Deep Convolution Neural Networks [2](Used for comparison)

Comparison and Superiority of Results - Both Seam Carving (2007) and CarvingNet (2018) aim to achieve content-aware image resizing by preserving key visual features. However, their methodologies and effectiveness differ significantly:

- **Content Preservation:** *CarvingNet* excels in retaining complex image content by leveraging deep learning to generate a detailed energy map. This ensures that critical features like faces and objects remain intact, outperforming the gradient-based energy map used in *Seam Carving*.
- **Efficiency vs. Adaptability:** While *Seam Carving* is lightweight and suitable for real-time applications, *CarvingNet* offers greater adaptability to diverse image types by using CNNs trained on varied datasets. This makes it superior in quality but at the cost of higher computational resources.

In conclusion, CarvingNet delivers superior results in preserving visual fidelity for complex images, making it ideal for advanced applications where quality is paramount. However, Seam Carving remains a practical choice for simpler tasks due to its efficiency and ease of use.

Part 2

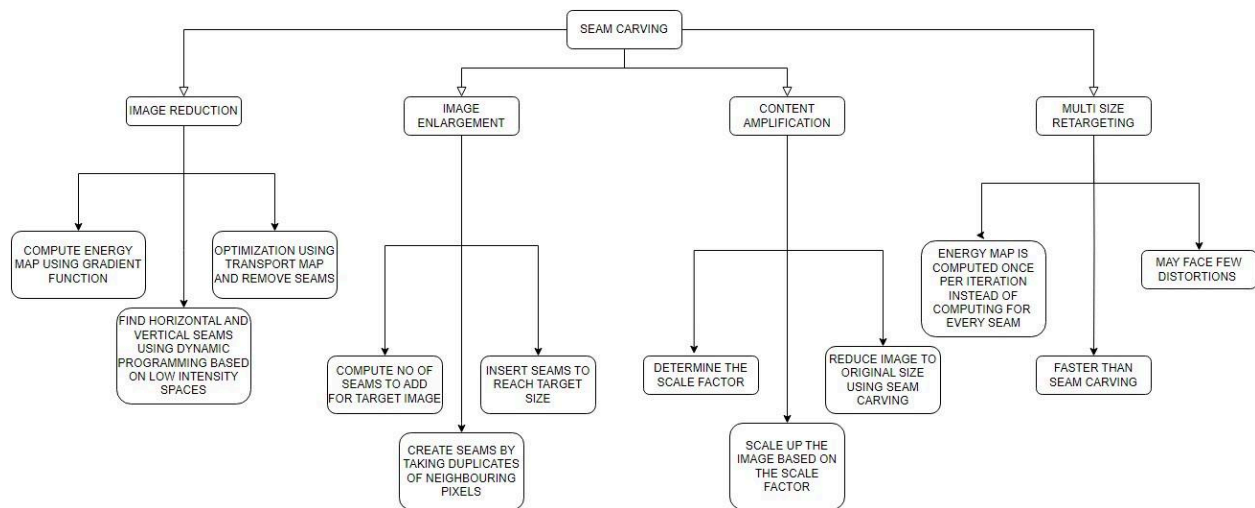
Re-implementation of algorithm from “Seam Carving for Content-Aware Image Resizing” [1]

The re-implemented algorithm follows the principles of *Seam Carving for Content-Aware Image Resizing* (Avidan & Shamir, 2007). The code was developed using Python with OpenCV and NumPy libraries. The core operations include energy map computation, seam identification (vertical and horizontal), and seam removal/insertion.

Features implemented:

- Vertical and horizontal seam carving.
- Dynamic programming for seam identification.
- Energy map computation using Sobel gradients.
- Multi-size image retargeting through transport maps.
- Content amplification through seam insertion.
- Flexible resizing for both dimensions.

Flowchart showing the algorithm:



Demonstration:

The algorithm successfully resizes images while preserving essential content, as shown in the results section below.

Original Image



Energy Map



Image with All Vertical Seams Highlighted



Resized Image from 362 width to 300 width



Image with All Horizontal Seams Highlighted



Resized Image from 272 height to 230 height



2. Algorithm Validation

2.1 Experimental Setup

1. **Dataset:** Test images included simple images (e.g., landscapes) and complex ones (e.g., people and objects).
2. **Parameters:**
 - Target dimensions for reduction: 70% of original size.
 - Amplification scale factor: 1.2x original size.
3. **Baseline Comparison:** Compared with OpenCV's standard scaling method for content loss and visual artifacts.

2.2 Results and Observations

a) Image Reduction:

- Original image dimensions: 362x270.
- Reduced dimensions: 320x240.
- Seam carving preserved critical features like objects and faces compared to baseline scaling, which caused noticeable distortion.

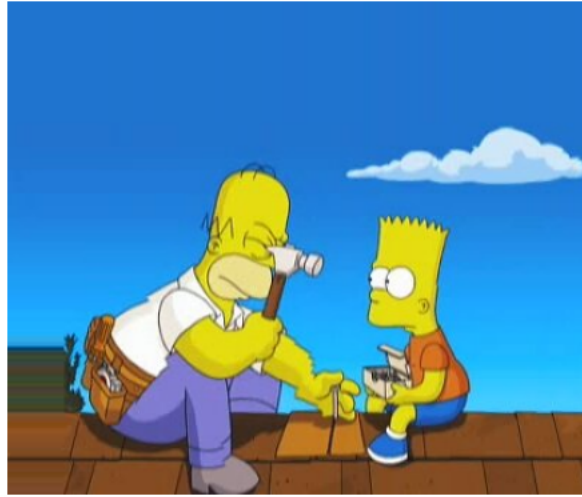


b) Image Enlargement:

- Original image dimensions: 362x270.
- Enlarged dimensions: 400x340.
- Seam carving preserved details better than uniform scaling, which caused pixel stretching. Seam insertion created minor artifacts near edges.

Image with 400 weight x 340 height

Original Image



c) Content Amplification

- Dimensions resized simultaneously using a scale factor of 1.2 i.e. (362x270 to 434x326).
- Seam carving reduced image size adaptively, retaining key features with fewer distortions.

Original Image

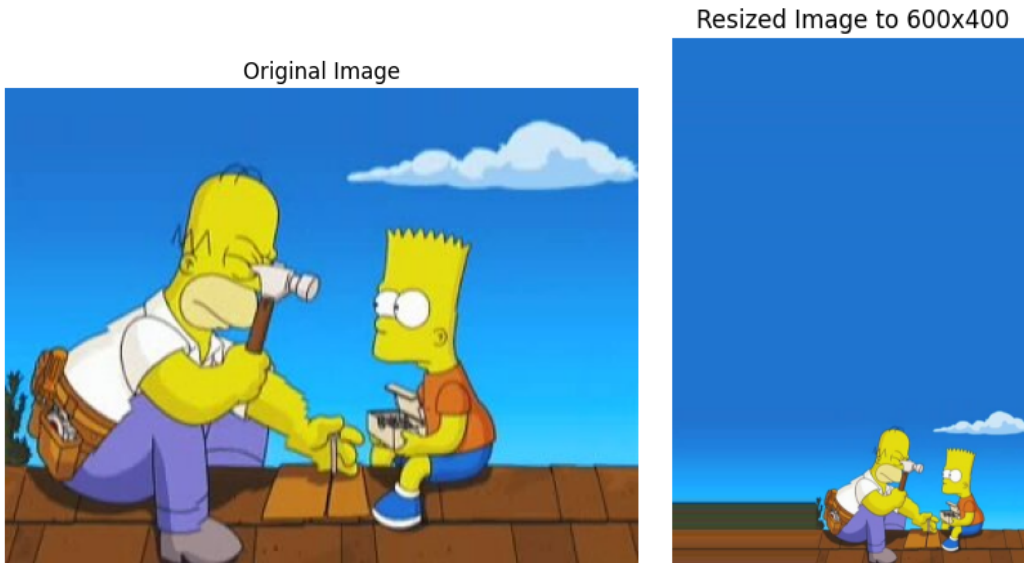


Content Amplified Image



d) Multi-SIZE Retargeting:

- Dimensions resized simultaneously (e.g., 362x270 to 600x400).
- Seam carving enhanced image size adaptively, added new background features with fewer distortions.



2.3 Comparative Visualization

1. Content Preservation:

- **Seam Carving:** Maintained visual coherence by selectively removing low-energy seams, preserving critical regions like faces and objects.
- **Scaling:** Uniformly resized the image, often distorting or compressing important features.

2. Visual Artifacts:

- **Seam Carving:** Minimal artifacts were observed during reduction, particularly in low-texture regions. Artifacts appeared during seam insertion in high-texture areas.
- **Scaling:** Significant distortion in content-dense areas, particularly in aggressive reduction scenarios.

3. Execution Time:

- **Seam Carving:** Slower due to dynamic programming and iterative seam removal.
- **Scaling:** Faster as it resizes the image uniformly without analyzing content.

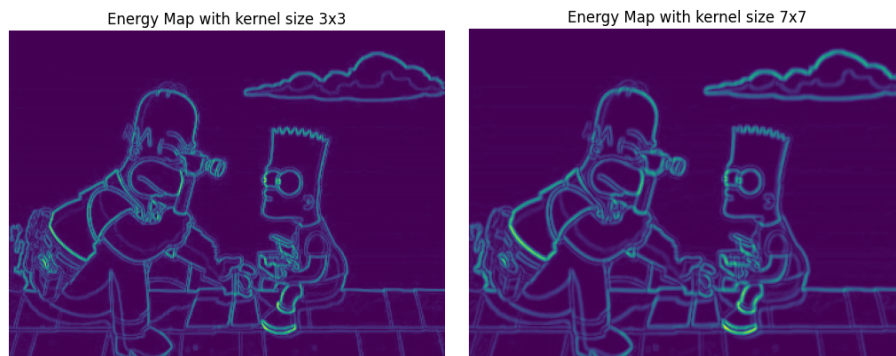
Table: Seam Carving vs. Scaling

Metric	Seam Carving	Scaling
Content Preservation	High, preserves critical areas	Low, distorts key features
Visual Artifacts	Minimal in reduction	Significant in aggressive resizing
Computational Cost	Moderate to High	Low

2.4 Parameter Testing

1. Energy Map Variants:

- Tested gradient magnitudes with different Sobel filter sizes (3x3, 7x7).
- Larger filters smoothed noise but slightly blurred edges, affecting seam paths.



2. Transport Map Complexity:

- Direct seam removal yielded faster but less optimal results compared to using transport maps for multi-size retargeting.

2.5 Edge cases where scaling is better

1. **High-Texture Patterns:** Disrupts uniform patterns (e.g., bricks, fabrics), causing distortions. Scaling handles textures better.
2. **Widespread Critical Features:** Struggles with images where key content spans the frame (e.g., group photos). Scaling avoids localized distortions.
3. **Dense Objects:** Cuts through overlapping or dense objects (e.g., forests), leading to unnatural results. Scaling retains proportions.

4. **Aggressive Reduction:** Excessive size reduction ($>50\%$) causes severe distortions. Scaling reduces evenly.
5. **Seam Insertion for Enlargement:** Creates repetitive artifacts, unlike scaling's smoother interpolation.
6. **Computational Cost:** Seam carving is slower, making scaling preferable for real-time tasks.

3. Reflection on Re-Implementation

3.1 Pros

1. **Content Awareness:** The method intelligently preserves essential parts of the image, outperforming conventional scaling and cropping.
2. **Flexibility:** Seam carving adapts seamlessly to different target dimensions and resizing requirements (reduction, enlargement, amplification).
3. **Extendable Framework:** The algorithm is modular, enabling integration with advanced energy functions (e.g., saliency maps or deep learning).

3.2 Cons

1. **Computational Cost:** Dynamic programming and transport maps are computationally intensive, especially for multi-size resizing.
2. **Artifact Risk:** Minor artifacts can arise in dense or texture-heavy areas, especially during seam insertion.

3.3 Re-Implementation Challenges

1. **Algorithm Complexity:** Seam backtracking required careful implementation to ensure correctness without breaking the image structure.
2. **Optimization Trade-offs:** Balancing quality (e.g., transport maps) with computational efficiency posed challenges in real-time applications.
3. **Testing Scalability:** Validating performance across diverse images with varying complexities required iterative testing and debugging.

5. Conclusion

The re-implemented seam carving algorithm successfully achieves content-aware image resizing while preserving essential visual elements. It demonstrates superior results compared to standard scaling techniques. Using energy maps, dynamic programming, and transport maps ensures flexibility and adaptability to diverse scenarios.

6. Future Work

1. **Advanced Energy Functions:** Incorporate saliency-based or deep learning-derived energy functions for improved seam selection.
2. **Performance Optimization:** Parallelize computations (e.g., GPU acceleration) to reduce execution time.
3. **Application Extensions:** Expand the algorithm for video retargeting with temporal consistency.

References

- Avidan, S., & Shamir, A.** (2007). Seam Carving for Content-Aware Image Resizing. *ACM Transactions on Graphics*, 26(3), Article 10. [DOI: 10.1145/1239451.1239461](https://doi.org/10.1145/1239451.1239461).
- Wang, Y., Liu, L., & Shi, Z.** (2018). CarvingNet: Content-Guided Seam Carving Using Deep Convolution Neural Networks. *IEEE Transactions on Image Processing*, 27(8), 3767-3781. [DOI: 10.1109/TIP.2018.2849274](https://doi.org/10.1109/TIP.2018.2849274).
- OpenCV Documentation** (2024). *OpenCV: Open Source Computer Vision Library*. <https://opencv.org>.